Problem Set 3

1.

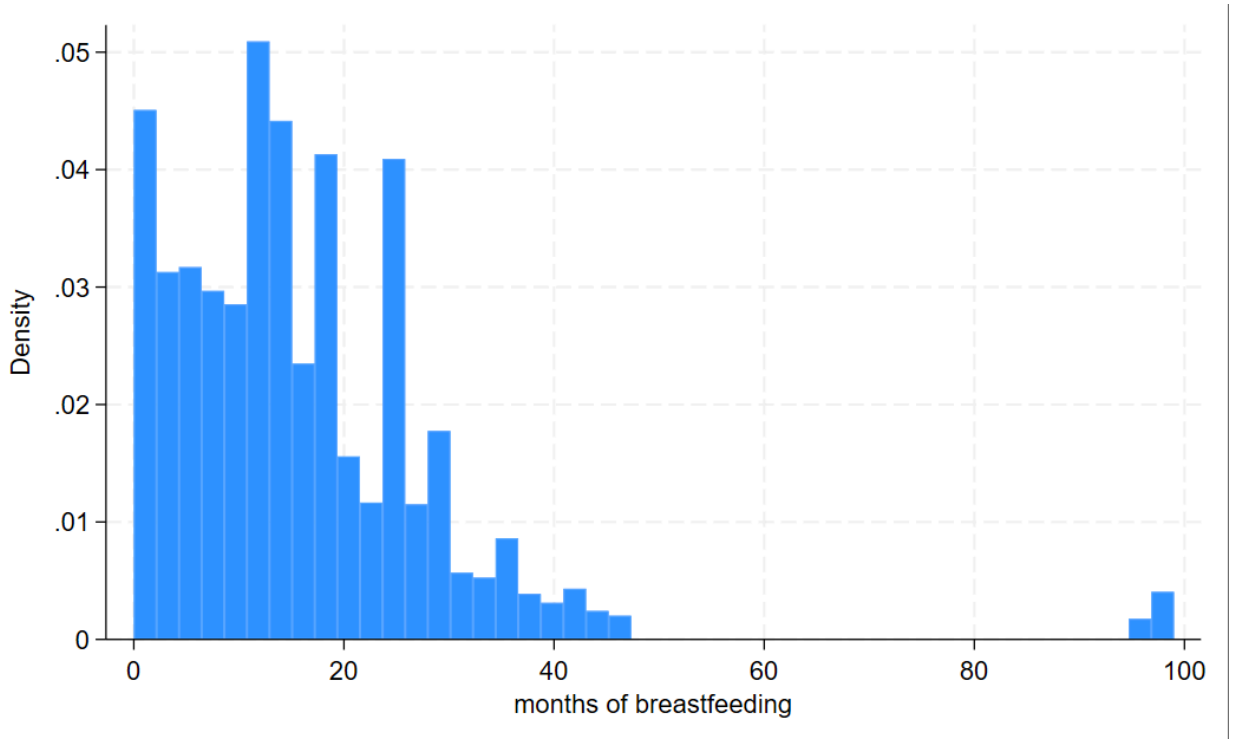
A.

Table 1: Descriptive Statistics

	Mean	SD	Observations
Male (0/1)	.514	.4998	48,959
Years of Education	3.095	4.3779	48,800
Electricity (0/1)	.5436	.4981	48,959
Died (0/1)	.0752	.2637	48,959
Age (years)	1.4622	1.125	45,279
Age (months)	22.8387	13.8716	45,275

Note: This table shows averages for women in India collected in 1992 and 1993. Age in years and months is only measured among children who did not die.

B.



What's unusual about this histogram is that there are extreme outliers that have allegedly been breastfeeding for just under 100 months. The cause of this might be data entry errors or a misunderstanding of the units (months) that are being used in this particular question. Maybe the participants thought they were asked days of breastfeeding instead.

—> <u>CORRECTION</u>: I cleaned the m5 variable, but not in the right way. I cleaned values that had some sort of phrase next to them like "dk", "no answer" but didn't clean the extra high numbers of 95 and 99. I thought they may have just been placeholders for mothers who breastfed past a certain point. When I filter those out, my histogram then matches the problem set solution and we see that there are clusters around 0 months, 12 months, 18 months, and 24 months. This could be due to a combination of mothers using those as a marker to intentionally stop breastfeeding and mothers using those numbers as their closest estimation when remembering the duration of breastfeeding.

According to our table for s504, 56.25 percent answered that the sex preference for their next child was male while just 14.62 percent answered that their preference for their next child was female. This indicates a strong preference towards male children. It is worth noting that about a third reported that it didn't matter or that the decision was up to God.

In the results for overall child preferences, we can see that a majority (roughly 58%) of women are indifferent to the sex of their child. But while many are indifferent, there is still 39% that have a preference for boys, leaving the preference for girls at about 3%. This is an interesting finding if we are trying to test if women often have a strong preference for boys.

—> <u>CORRECTION:</u> Again, I need to do a better job of cleaning my variables. I cleaned the s517 variables for values of 99 but did not clear them for ALL values above 90. I need to make a habit of tabulating each variable and inspecting each high value before I actually use it in a regression or calculation. This is the cause for my percentages being slightly different than the problem set solutions.

2.

A. B_i = duration of breastfeeding M_i = Male child dummy variable e = error term
   B_i = alpha + beta * M_i + e

—> <u>CORRECTION:</u> Add an i to the error term. This is because the error term applies to each individual.

B. Table below

Table 1: Relationship between gender of a child and the duration of breastfeeding, no controls

Male Child	.5766
Constant	15.9766
Observations	44,478
R-squared	.0005

Note: This table shows the relationship for women in India collected in 1992 and 1993. Children who died were not included in this regression.

The constant: among women in India whose child did not die, the average duration of breastfeeding for their female child is 15.9766 months.
The beta: among women in India whose child did not die, the average duration of breastfeeding when their child is male is .5766 months longer
However, since our p-value is very very small, we know that there is a statistically significant difference between breastfeeding durations among girls and boys.

C. Table below

Table 2: Relationship between gender of a child and the duration of breastfeeding, with controls

Male child (0/1)	.5716	.5821
Years of mother's education	-.3355	-.2707
Has electricity (0/1)		-1.4507
Constant	17.0398	17.63198
Observations	44,333	44,333
R-squared	.0143	.0170

Note: This data comes from women in India, collected in 1992 and 1993. Children who died were not included in this regression.

Looking at regression 2, column 2: Holding all else constant, for each additional year of education earned by the mother, the duration of breastfeeding is expected to decrease by .2707 months.

Looking again at regression 2: holding all else constant, having electricity is associated with a 1.4507 month reduction in the duration of breastfeeding

Comparing the two columns, we see that the coefficient on years of mother's education is weaker. This is likely because we controlled for electricity. Before, the impacts of having electricity were embedded in our education variable. Now, we take that out and give it its own place in the regression equation, weakening the education variable as a result.

D. Table below

Table 3: Relationship between gender of a child and the duration of breastfeeding by gender preference

	Prefer boys	Prefer girls	Indifferent
Male child (0/1)	.5326	-.2738	.5384
Constant	16.4856	15.8073	15.6926
Observations	17,321	1,337	25,820
R-squared	.0004	.0001	.0005

Note: This data comes from women in India, collected in 1992 and 1993. Children who died were not included in these regressions.

From this table, it becomes a little more clear that women who prefer boys will breastfeed their son longer than a woman who prefers girls will breastfeed their daughter. The difference is seen in the absolute values of the Male child row.

→ **<u>CORRECTIONS:</u>** Even after the lab, I was unsure of the formula for generating new dummy variables. Instead of checking the packet again, I searched for the formula online. That formula ended up giving me slightly different numbers for the entirety of question 2. I think the biggest point of confusion here was understanding that the variable is already cleaned through your formula. For example, “gen male = 1 if b4 == 1” and “replace male = 0 if b4 == 2” not only creates your new variable but omits any missing values too. I went back into Stata to practice the right commands and pasted one variable example below. Now, my numbers look more accurate.

History

Filter commands here

#	Command	_rc
1	use "C:\Users\broad\Downloads\IA...	
2	gen child_died =1 if b6!=.	
3	replace child_died =0 if b6==.	
4	replace m5 if m5 == 94	100
5	replace m5 =0 if m5 ==94	
6	replace m5 =. if m5> 94	
7	reg m5 male if child_died == 0	111
8	gen male =1 if b4==1	
9	replace male = 0 if b4 ==2	
10	reg m5 male if child_died = 0	198
11	reg m5 male if child_died ==0	
12	tab male	
13	tab m5	

```
. replace m5 =. if m5> 94
(619 real changes made, 619 to missing)

. reg m5 male if child_died == 0
variable male not found
r(111);

. gen male =1 if b4==1
(23,796 missing values generated)

. replace male = 0 if b4 ==2
(23,796 real changes made)

. reg m5 male if child_died = 0
invalid syntax
r(198);

. reg m5 male if child_died ==0

      Source |       SS           df       MS      Number of obs   =    44,843
-------------+----------------------------------   F(1, 44841)     =     41.51
       Model |  4266.44822         1  4266.44822   Prob > F        =    0.0000
    Residual |  4609214.33    44,841  102.790177   R-squared       =    0.0009
-------------+----------------------------------   Adj R-squared   =    0.0009
       Total |  4613480.77    44,842  102.883029   Root MSE        =    10.139

------------------------------------------------------------------------------
          m5 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        male |   .6171164   .0957877     6.44   0.000     .4293709    .8048619
       _cons |     14.947   .0686197   217.82   0.000      14.8125     15.0815
------------------------------------------------------------------------------
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3.

A. I want to test whether or not the sex of the household head has any impact on whether you watch tv every week. The null hypothesis is that there is no difference between female heads and male heads. My personal hypothesis is that having a male household head increases the probability that you watch tv every week

B. T_i = alpha + beta * M_i + e
Where T_i is a dummy variable noting if you watch tv every week or not and M_i is a dummy variable noting if the sex of the household head is male or not.

C. Screenshot of regression below

```
. reg v110 v151

      Source |       SS           df       MS      Number of obs   =    48,936
-------------+----------------------------------   F(1, 48934)     =    156.76
       Model |  33.2215159         1  33.2215159   Prob > F        =    0.0000
    Residual |  10370.0971    48,934  .211920078   R-squared       =    0.0032
-------------+----------------------------------   Adj R-squared   =    0.0032
       Total |  10403.3186    48,935  .212594638   Root MSE        =    .46035

------------------------------------------------------------------------------
        v110 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        v151 |   .1021396   .0081578    12.52   0.000     .0861503    .1181289
       _cons |   .1972979   .0089732    21.99   0.000     .1797104    .2148855
------------------------------------------------------------------------------
```

Results: We see that we have an extreme t value and a very small P value. This tells us that we can reject the null hypothesis that there is no difference. We now see that if the head of your household is male, the probability that you watch tv every week increases by 10 percentage points.